

LINDI SANG

Email: e1553738@u.nus.edu

Mobile: +65 80994812

RESEARCH INTERESTS

- Deep Learning-driven Materials Discovery, AI Infrastructure, Computational Methods, and Multiscale Simulation
- Others: Hardware-Software Co-design for Edge AI, Semiconductor Technology

EDUCATION

National University of Singapore

Master of Science in Materials Science and Engineering

- GPA: 4.75/5

Singapore

Aug 2025 - Present

National University of Singapore (Suzhou) Research Institute

Exchange Student - Undergraduate Final Year Project

Suzhou, China

Aug 2024 - May 2025

University of Electronic Science and Technology of China (UESTC)

Bachelor of Engineering in Materials Science and Engineering

- GPA: 3.9/4.0 - Rank: 2/21

Chengdu, China

Aug 2021 - Jun 2025

HONOR & AWARDS

- Excellent Competitor in HiSilicon Embedded Competition, HiSilicon, 2024
- EVE ENERGY Enterprise Scholarship, EVE, 2024
- Far East Enterprise Scholarship, Far East, 2024
- Model Student Scholarship, UESTC, 2024
- Excellent Student (Second Class) Scholarship, UESTC, 2023
- Model Student Scholarship, UESTC, 2022

COMPETITIONS

- Third Prize in the Finals, 2024 National College Students Embedded Chip and System Design Competition (SoC China)
- Honored with First Prize in Central and Southwestern China Division, 2024 HUAWEI CUP National Undergraduate IoT Design Contest
- First Prize, 2023 Sichuan International College Students' Innovation Competition

INTERNSHIPS

A*STAR Institute of High Performance Computing (A*STAR IHPC)

Master Intern - NUS MLE5003 Materials Science & Engineering Project

Aug 2025 - Present

EVE ENERGY Co., Ltd (Huizhou, China)

Intern, Battery Management System Department

Jan 2024

RESEARCH EXPERIENCE

AI-Accelerated First-Principles Calculations for Catalyst Discovery by Leveraging Graph Neural Networks

NUS MLE5003 Project

Aug 2025 - Present

- **Model Fine-tuning:** Fine-tuned the Universal Model for Atoms (UMA) on the spin-polarized AQCat25 dataset using a Mixture of Linear Experts (MOLE) architecture to enhance predictive accuracy for catalytic interfaces.
- **Workflow Automation:** Developed the NRR-Explorer Agent integrating AdsorbML to autonomously perform site sampling, geometry relaxation, and structural anomaly filtering.
- **Catalyst Discovery:** High-Throughput Screening to identify optimal reaction pathways and pinpoint high-performance catalyst candidates

Bio-realistic Memristive Synapse for Neuromorphic Computing

Final Year Project

Sep 2024 - May 2025

- **Device Fabrication:** Fabricated Mo/AlScN/Pt memristors on SiO₂ substrates using magnetron sputtering and photolithography in a cleanroom environment.
- **Characterization & Testing:** Analyzed ferroelectric switching and resistive switching mechanisms via XRD and PUND
- **Performance Metrics:** Achieved optimized device performance including a high ON/OFF ratio 10³, long-term retention 10⁵s, and nanosecond-level synaptic weight update speeds.

Spin-dependent Transport in Binuclear Transition Metallic Phthalocyanine-based Spintronic Device

College Student Innovation Training Program

Dec 2022 - Sep 2023

- **Quantum Transport Modeling:** Designed molecular spintronic devices using Device Studio, bridging half-metallic 1D-AGNR-FeN₄ nanoribbons with binuclear transition-metal-filled phthalocyanine molecules.
- **First-principles Simulation:** Performed structural optimization via VASP and conducted quantum transport simulations via Nanocal based on the NEGF-DFT framework to investigate spin-polarized electron transport.

Theoretical Analysis and Simulation of M-ary Modulation Schemes over AWGN Channels

NUS EES 4402 Project

Dec 2024

- Derived BER vs. SNR expressions for 2-PAM, 4-PAM, and QPSK schemes over AWGN channels using ML detection.
- Analyzed the theoretical BER bounds and simulated baseband transmission, achieving a high-precision BER resolution of 10^{-7} .
- Compared the transmission efficiency of various modulation schemes.

Intelligent Food Storage System Based on HiSilicon Hi3861 & Hi3516

HiSilicon Embedded Competition

Dec 2023 - Aug 2024

- **Hardware Design:** Developed an IoT system based on HiSilicon Hi3861 for multi-sensor data collection and low-power wireless communication.
- **Computer Vision (CV) Integration:** Deployed and quantized YOLOv5 models onto the HiSilicon Hi3516 board, optimizing edge-side inference for real-time item recognition.
- **Full-Stack App:** Designed a WeChat Mini-program to interface with cloud data, providing real-time monitoring and intelligent storage recommendations.

SKILLS

- **Software & Infrastructure:** Python and MATLAB for scientific computing; LaTeX for writing; PyTorch for machine learning workflows; Proficiency in Linux and High-Performance Computing (HPC).
- **Digital & Embedded Systems:** Verilog and FPGA design; C/Cpython for FreeRTOS and Embedded Linux development; PCB Layout.
- **Multiscale Simulation:** VASP and Nanocal (DFT/NEGF); Ansys Fluent (CFD).
- **Others:** DOE (Design of Experiments) and statistical data analysis via Minitab; Strong foundation in Solid State Physics and Semiconductor Device Theory.